

Integral Divisor Products on the Nonstandard A_5 -Invariant Cubic Pencil

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Abstract

We study the nonstandard A_5 -invariant pencil of smooth cubic threefolds. A six-axis polarization identifies its exotic two-primary gluing packet and constrains odd-degree product decompositions of the intermediate Jacobian. An all-degree integral graph-saturation theorem then proves algebraicity of the primitive minimal class for every smooth member. Consequently every member is universally CH_0 -trivial. The family is non-isotrivial, and all but its Fermat point lie outside both the separated-variable locus and the explicit coprime-degree family of Yang–Yu–Zhu. Combined with the one-stabilization theorem for cubic threefolds, this gives a non-isotrivial family whose products with $\mathbf{P}^1$ are universally CH_0 -trivial and irrational.

Keywords. cubic threefold; A_5 -action; universal CH_0 -triviality; intermediate Jacobian; minimal class; integral divisor products.

1 Introduction

Let W_5 be the five-dimensional irreducible representation of A_5 . The invariant space

$$(\mathrm{Sym}^3 W_5^*)^{A_5}$$

has dimension two, so its projectivization is a pencil of cubic threefolds. We prove universal CH_0 -triviality uniformly over its smooth locus by constructing the primitive minimal class of the intermediate Jacobian.

Theorem 1.1 (The nonstandard A_5 -pencil). *Let B° be the smooth locus in $\mathbf{P}((\mathrm{Sym}^3 W_5^*)^{A_5})$, and let $\mathcal{X} \rightarrow B^\circ$ be the resulting family. For every $b \in B^\circ(\mathbf{C})$,*

$$\mathcal{X}_b \quad \text{and} \quad \mathcal{X}_b \times \mathbf{P}^1$$

are universally CH_0 -trivial, while $\mathcal{X}_b \times \mathbf{P}^1$ is irrational. The family is non-isotrivial in coarse moduli. Exactly one point of its coarse-moduli image is represented by a cubic whose defining form is a sum of cubic forms in pairwise disjoint nonempty groups of at most three variables: the Fermat point.

The geometric input is a six-axis configuration on the Fano surface of lines. Its polarization matrix has a unit direction and a four-dimensional depth-one part. The principal gluing kernel is selected by a quadratic two-primary orientation, and its integral graph coordinates satisfy a rank-one generation theorem in every degree. Voisin’s criterion then turns the resulting algebraic primitive minimal class into universal CH_0 -triviality. Irrationality after one stabilization is imported from [18]; it is logically independent of the cycle construction here.

The character of W_5 is $(5, 1, -1, 0, 0)$ on the classes $1, 2A, 3A, 5A, 5B$. The symmetric-cube character formula gives

$$\dim(\mathrm{Sym}^3 W_5^*)^{A_5} = 2.$$

Hartlieb identifies the resulting coarse-moduli image with his M_{H_1} , a one-dimensional special subvariety [7, Lemma 5.5 and Proposition 5.7].

Remark 1.2. The pencil contains the Fermat cubic. The representation W_5 is induced from a nontrivial cubic character of a point stabilizer $A_4 \subset A_5$; in a monomial realization, every nonzero matrix entry is a cube root of unity, so $x_1^3 + \cdots + x_5^3$ is invariant. This is the unique separated-variable point in the coarse-moduli image.

Section 2 determines the polarization and gluing packet. Section 3 proves the all-degree integral saturation theorem and its universal- CH_0 consequence. The final section proves Theorem 1.1 and records the relation with the one-stabilization theorem.

2 The six-axis polarization and the exotic packet

Let Ω be the six conjugate D_5 -subgroups of A_5 , equivalently the six Sylow-5 subgroups with their normalizers, and put

$$H_p = \mathrm{Aug}(\mathbf{F}_p^\Omega) / \langle \mathbf{1} \rangle \quad (p = 2, 3).$$

We record the modular representation calculation used throughout the six-axis argument, so that the commutants do not have to be inferred from a table.

Lemma 2.1 (The six-point hearts). *For $p = 2, 3$, the $\mathbf{F}_p A_5$ -module H_p is simple, and*

$$\mathrm{End}_{A_5}(H_2) = \mathbf{F}_4, \quad \mathrm{End}_{A_5}(H_3) = \mathbf{F}_3.$$

Proof. Identify the six-point action with that of $A_5 \simeq \mathrm{PSL}_2(\mathbf{F}_5)$ on $\mathbf{P}^1(\mathbf{F}_5)$, whose points are $\infty, 0, 1, 2, 3, 4$; the stabilizer of a point is the normalizer of a Sylow-5 subgroup. Let $a(z) = z + 1$ and $s(z) = -1/z$, which generate $\mathrm{PSL}_2(\mathbf{F}_5)$. In $H_p = \mathrm{Aug}(\mathbf{F}_p^\Omega) / \langle \mathbf{1} \rangle$, use the classes of

$$e_\infty - e_4, \quad e_0 - e_4, \quad e_1 - e_4, \quad e_2 - e_4$$

as a basis. The two generators act, for $p = 2$, by

$$A_2 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

and, for $p = 3$, by

$$A_3 = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{pmatrix}, \quad S_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 2 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

In both characteristics A_p has minimal polynomial $\Phi_5(t) = t^4 + t^3 + t^2 + t + 1$, which is irreducible over $\mathbf{F}_p$ because the order of p in $(\mathbf{Z}/5\mathbf{Z})^\times$ is four. Hence any nonzero A_5 -stable subspace, being A_p -stable, is all of H_p ; thus H_p is simple.

It remains to compute the commutant. Solving the linear equations $XA_p = A_pX$ and $XS_p = S_pX$ gives only scalar matrices for $p = 3$. For $p = 2$ it gives the two-dimensional algebra spanned by I and

$$T = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \end{pmatrix}, \quad T^2 + T + I = 0.$$

Thus the two commutants are $\mathbf{F}_3$ and $\mathbf{F}_2[T]/(T^2 + T + 1) = \mathbf{F}_4$, respectively. This agrees with the modular matrix tables in [8]. $\square$

The six axes arise from the Fano surface. Let S be the Fano surface of a smooth member of the A_5 -pencil and $(J, \Theta) = \text{Alb}(S)$ its intermediate Jacobian. Each of the six D_5 -subgroups gives an elliptic quotient $q_H : J \rightarrow E_H^{\text{quot}}$ and its Rosati dual $i_H^{\text{quot}} : E_H^{\text{quot}} \rightarrow J$. **Generic non-CM axis.** Hartlieb's elliptic isogeny factor, and hence every six-axis elliptic quotient, is non-CM at the geometric generic point of the pencil. Indeed, a CM endomorphism has a fixed imaginary-quadratic order, so the generic j -invariant is a root of its Hilbert class polynomial and is constant. After a finite base change the elliptic factor would then be isotrivial. Hartlieb's isogeny $J_b \sim E_b^5$ would put the whole intermediate-Jacobian family in the countable isogeny class of one fixed abelian variety. The image of an algebraic curve in a moduli space cannot be both positive-dimensional and countable: its image is constructible, and a positive-dimensional constructible subset contains a nonempty Zariski-open subset of an algebraic curve, hence has uncountably many complex points. This contradicts the fact that the closure of the intermediate-Jacobian image is a one-dimensional special subvariety [7, Proposition 5.7 and Remark 5.8]. Since the generic intermediate Jacobian is isotypic to E_b^5 , each nonzero elliptic quotient is isogenous to E_b , and so is non-CM as well. This is a statement about the geometric generic point, not about every member: individual fibres may have complex multiplication, and there the divisor lattice selected by the marked presentation can be smaller than the full Neron–Severi lattice. Those fibres are covered by the prescribed-lattice reading of Theorem 3.3 recorded after its proof.

Proposition 2.2 (Generic six-axis polarization). *On the geometric generic fibre of the nonstandard A_5 -pencil, after coherent identification of the six quotient elliptic curves, the Rosati Gram matrix of the six maps is*

$$G_6 = 6I_6 - J_6.$$

Any five axes therefore give a generic-fibre isogeny $E^5 \rightarrow J$ whose pulled-back polarization has Smith type $(1, 6, 6, 6, 6)$.

Proof. The geometric generic elliptic factor has only integer endomorphisms. We recall the numerical passage from Roulleau's genus-two configuration to the six elliptic fibres. Let $D_g \subset S$ be the curve indexed by an involution $g \in A_5$, put $D_{\text{tot}} = \sum_g D_g$, and, for a dihedral subgroup H of order ten, put

$$F_H = \sum_{g \in H, g^2=1} D_g.$$

Roulleau proves that F_H is a fibre of $S \rightarrow E_H^{\text{quot}}$ [11, Theorem 11(D)], so $F_H^2 = 0$. His intersection table has $D_g^2 = -4$, and distinct curves meet with multiplicity zero, two, or one when the product of their labels has order two, three, or five, respectively; Lemma 14 proves

that these values are constant in the corresponding families [11, Lemma 14, Corollary 16, and (3.1)]. Among the 105 unordered pairs of involutions, the product orders two, three, and five occur 15, 30, 60 times. Hence

$$D_{\text{tot}}^2 = 15(-4) + 2(30 \cdot 2 + 60 \cdot 1) = 180.$$

Every involution belongs to two of the six dihedral subgroups, and therefore $\sum_H F_H = 2D_{\text{tot}}$. Two-transitivity on the six subgroups now gives

$$30(F_H \cdot F_{H'}) = (2D_{\text{tot}})^2 = 720, \quad F_H \cdot F_{H'} = 24 \quad (H \neq H').$$

By the preceding generic non-CM argument, the elliptic factor has only integer endomorphisms. After coherent identification of the generic elliptic quotients, two-transitivity makes the Rosati matrix have diagonal d and off-diagonal m . Put

$$D_H = q_H^*[0] \in \text{NS}(J), \quad N_H = i_H^{\text{quot}} q_H \in \text{End}(J).$$

Then $D_H|_S = F_H$, and N_H is the Rosati endomorphism attached to D_H . Coherent identification gives $q_H i_H^{\text{quot}} = [d]$ and $q_H i_{H'}^{\text{quot}} = [m]$ for $H \neq H'$. Each target identification is determined only up to a sign, and the ambiguity changes nothing used here. Replacing the chosen isomorphism $E_H^{\text{quot}} \rightarrow E$ by its negative replaces q_H by $-q_H$ and, by functoriality of Rosati duality, i_H^{quot} by $-i_H^{\text{quot}}$ at the same time. Hence $N_H = i_H^{\text{quot}} q_H$ is unchanged, and so is $D_H = q_H^*[0]$, since $[-1]^*[0] = [0]$. The diagonal entry $q_H i_H^{\text{quot}}$ is likewise unchanged, while the Gram matrix changes by a signed congruence DG_6D with $D = \text{diag}(\pm 1)$, which alters neither the Smith type of the pulled-back polarization nor anything below. The coherent choice is the one making all off-diagonal entries equal; it is unique up to a global sign, which acts trivially. The invariant sum of the six axis maps vanishes because the five-dimensional constituent has no trivial summand; composing with q_H gives $d + 5m = 0$. The minimal-class identity $[S] = \Theta^3/3!$ [3], together with the polarized Riemann–Roch trace formula

$$\frac{\alpha\beta\Theta^{g-2}}{(g-2)!} = \text{tr}(N_\alpha) \text{tr}(N_\beta) - \text{tr}(N_\alpha N_\beta),$$

obtained by polarizing the reduced-norm Riemann–Roch polynomial [6, Theorem 4.1], gives the following identities. Here tr is the degree-five trace in the coefficient representation, the trace entering that reduced norm, rather than the degree-ten trace on rational first homology:

$$\text{tr}(N_H) = d, \quad \text{tr}(N_H N_{H'}) = m^2, \quad F_H F_{H'} = d^2 - m^2.$$

Thus $d^2 - m^2 = 24$. Together with $d + 5m = 0$ and $d > 0$, this forces $(d, m) = (5, -1)$, hence $G_6 = 6I - J$. A five-by-five minor has Smith form $(1, 6, 6, 6, 6)$. $\square$

Let B° be the connected smooth locus of the nonstandard A_5 -pencil, and let $\mathcal{J} \rightarrow B^\circ$ be its relative intermediate Jacobian.

Lemma 2.3 (Relative six-axis source). *There are an elliptic scheme $\mathcal{E} \rightarrow B^\circ$ and a relative isogeny*

$$f : \mathcal{A} := \mathcal{E} \otimes_{\mathbf{Z}} (\mathbf{Z}^\Omega / \mathbf{Z}\mathbf{1}) \longrightarrow \mathcal{J}$$

which is A_5 -equivariant. The quotient lattice carries the form induced by $6I_6 - J_6$; both the source and this form carry their natural S_6 -action. Moreover,

$$f^* \lambda_\Theta = \lambda_{\mathcal{A}}. \tag{2.1}$$

After omitting one axis, $\mathcal{A} \simeq \mathcal{E}^5$ and this form has matrix $6I_5 - J_5$. For $p = 2, 3$, put $\mathcal{V}_p = \mathcal{E}[p]$. The p -primary kernel of the source polarization has a natural A_5 -equivariant symplectic description

$$\mathcal{D}_p \simeq H_p \otimes_{\mathbf{F}_p} \mathcal{V}_p \quad (p = 2, 3). \quad (2.2)$$

The p -primary part $\mathcal{K}_p$ of $\ker f$ is an A_5 -stable relative maximal isotropic subgroup of $\mathcal{D}_p$.

Proof. For $H \in \Omega$, the integral norm $n_H = \sum_{h \in H} h$ is a relative endomorphism of $\mathcal{J}$. Its action on rational first homology is ten times the projector onto the H -fixed summand. Fibrewise, Hartlieb's character calculation and isogeny decomposition give $H^1(J, \mathbf{Q}) \simeq W_5 \otimes H^1(E, \mathbf{Q})$, and hence the self-product of an elliptic curve through the five-dimensional module W_5 [7, Lemma 5.5 and Remark 5.8]. Since $\dim W_5^H = 1$, the fixed summand has rank two. More concretely, $n_H^2 = 10n_H$. The scheme-theoretic image of a homomorphism of abelian schemes in characteristic zero is an abelian subscheme, and its quotient factorization commutes with base change [1, Theorem A]. That citation supplies the abelian-scheme structure; the relative dimension is pinned by $n_H^2 = 10n_H$, which makes $n_H/10$ a projector, so on every fibre the image is the connected abelian subvariety it cuts out, of dimension one by $\dim W_5^H = 1$. Thus the scheme-theoretic norm image $\mathcal{E}_H^{\text{im}} \subset \mathcal{J}$ is an elliptic scheme. Since $hn_H = n_H$ for every $h \in H$, the subgroup H acts trivially on $\mathcal{E}_H^{\text{im}}$. Moreover $N_{A_5}(H) = H$, so transport by an element carrying H to H' is independent of the chosen transporter. The six norm-image elliptic schemes are therefore coherently isomorphic; fix one of them as $\mathcal{E}$.

Write $i_H^{\text{im}} : \mathcal{E} \rightarrow \mathcal{J}$ for the transported primitive norm-image inclusions. On the geometric generic fibre, Hartlieb's isogeny decomposition identifies $\text{Hom}^0(E, J)$ with the irreducible module W_5 . The coherently transported maps $(i_H^{\text{im}})_{H \in \Omega}$ form one A_5 -orbit, so their sum is invariant. Since $W_5^{A_5} = 0$, that sum is zero. The connected smooth curve B° is integral, $\mathcal{E}$ is integral, and $\mathcal{J}$ is separated over B° . Two morphisms from an integral scheme to a separated one agree along a closed subscheme; here that subscheme contains the dense generic fibre, so $\sum_H i_H^{\text{im}}$ and the zero map agree everywhere and $\sum_H i_H^{\text{im}} = 0$ over B° . Hence $(i_H^{\text{im}})_{H \in \Omega}$ descends from $\mathcal{E}^\Omega$ to the displayed map $f : \mathcal{A} \rightarrow \mathcal{J}$, and the construction is A_5 -equivariant.

Roulleau constructs the D_5 -fibration from an Albanese quotient $q_H : J \rightarrow E_H^{\text{quot}}$ by a connected four-dimensional abelian subvariety [11, the sentence following Lemma 17 and the proof of Theorem 11(D)]. Thus $F_H = q_H^*[0]_S$. The connected kernel makes the Rosati dual of q_H a primitive inclusion, and its image is the unique H -fixed elliptic subvariety. It therefore agrees, up to an automorphism of the target elliptic curve, with the norm-image elliptic curve $\mathcal{E}_H^{\text{im}} \hookrightarrow \mathcal{J}$. This also identifies the norm-image axis, up to isogeny, with Hartlieb's generic elliptic isogeny factor. The quotient curve E_H^{quot} and the norm-image curve $\mathcal{E}_H^{\text{im}}$ are therefore kept distinct until this point; the Rosati dual identifies the latter with the image of the former, up to the target automorphism just described. Consequently the generic-fibre classes computed in Proposition 2.2 are the classes of these relative homomorphisms. They give diagonal entry 5 and off-diagonal entry -1 on the geometric generic fibre. Applying the same generic-density and separatedness argument to these homomorphisms gives the identities over B° , hence (2.1). Choosing any five coordinate classes identifies the quotient lattice with $\mathbf{Z}^5$, where its matrix is $6I_5 - J_5$. Since this has Smith type $(1, 6, 6, 6, 6)$, the source polarization is nondegenerate, so f is a finite flat isogeny. From $\lambda_{\mathcal{A}} = f^\vee \lambda_\Theta f$ one gets

$$|\ker \lambda_{\mathcal{A}}| = 6^8, \quad |\ker f| = \deg f = 6^4.$$

For $x, y \in \ker f$, functoriality of the polarization commutator pairing gives $e_{\lambda_{\mathcal{A}}}(x, y) =$

$e_{\lambda_{\Theta}}(fx, fy) = 1$. Thus $\ker f$ is isotropic and has square-root order in $\ker \lambda_{\mathcal{A}}$, hence is maximal isotropic. Its p -primary part has fibrewise order p^4 for $p = 2, 3$.

For $p = 2, 3$, the primary discriminant local system is

$$\mathcal{D}_p = \ker(6I_6 - J_6 : \mathcal{A}[p] \longrightarrow \mathcal{A}^{\vee}[p]).$$

Modulo p , the radical quotient of the induced coefficient form is canonically

$$\text{Aug}(\mathbf{F}_p^{\Omega})/\langle \mathbf{1} \rangle = H_p.$$

The form on this coefficient discriminant is the normalized form

$$b_p(\bar{x}, \bar{y}) = \frac{1}{p} x^{\dagger} (6I_6 - J_6) y \pmod{p} = \begin{cases} \bar{x} \cdot \bar{y}, & p = 2, \\ -\bar{x} \cdot \bar{y}, & p = 3. \end{cases} \quad (2.3)$$

For augmentation lifts, the product of the coordinate sums is divisible by p^2 , which proves the displayed formulas. They are unchanged by adding the constant vector or p times an integral vector. The dot product on the augmentation module has radical exactly the constant line, so b_p is nondegenerate on H_p ; b_2 is alternating and b_3 is symmetric. Tensoring with the Weil pairing on $\mathcal{V}_p$ gives (2.2) as a symplectic isomorphism. Since f is A_5 -equivariant, each $\mathcal{K}_p$ is A_5 -stable. The preceding order and isotropy calculation makes it a relative maximal isotropic subgroup of this rank-eight local system. $\square$

Consequently the generic Gram calculation of Proposition 2.2 extends, by the relative homomorphisms and separatedness in Lemma 2.3, to every smooth fibre of the pencil. Thus every smooth fibre has six-axis Rosati matrix $6I_6 - J_6$, and any five axes have Smith type $(1, 6, 6, 6, 6)$.

Proposition 2.4 (Principal gluing packet). *The A_5 -stable two-primary principal kernels of the six-axis source are parametrized by*

$$\mathbf{P}^1(\mathbf{F}_4) = \mathbf{P}^1(\mathbf{F}_2) \sqcup \{\omega, \omega^2\}.$$

The three $\mathbf{F}_2$ -rational kernels are S_6 -stable. Each of the other two has stabilizer exactly A_5 , and the outer normalizer exchanges them. The intermediate Jacobian of the nonstandard A_5 -cubic component uses the latter pair. In the relative packet, monodromy on this exotic pair is the sign of its permutation action on the three rational kernels. Thus the exotic pair represents the orientation, or discriminant, class of the rational degree-three cover. The relative kernel selects one exotic sheet, so the mod-two monodromy of $\mathcal{E}$ is contained in $A_3 \simeq C_3$. The A_5 -stable three-primary kernels form $\mathbf{P}^1(\mathbf{F}_3)$; after choosing a complementary symplectic ruling, each is a scalar graph.

Proof. At two, by Lemma 2.1, the coefficient discriminant is the simple four-dimensional $\mathbf{F}_2 A_5$ -module H_2 , whose commutant is $\mathbf{F}_4$. The full discriminant is the semisimple module $H_2 \otimes_{\mathbf{F}_2} \mathbf{F}_2^2 \simeq H_2^{\oplus 2}$. Every stable half is therefore one copy of H_2 , and Schur's lemma identifies such submodules with the one-dimensional $\mathbf{F}_4$ -subspaces of $\mathbf{F}_4^2$. This gives $\mathbf{P}^1(\mathbf{F}_4)$.

Here is the polarization check explicitly. Identify the coefficient heart with the natural two-dimensional $\mathbf{F}_4$ -module. The trace-determinant form is a nonzero invariant alternating form. Relative to it, the normalized form b_2 of (2.3) differs by a nonzero scalar in $\mathbf{F}_4$; rescaling an $\mathbf{F}_4$ -basis absorbs that scalar because squaring permutes $\mathbf{F}_4^{\times}$. We may therefore put

$$b(x, y) = \text{Tr}_{\mathbf{F}_4/\mathbf{F}_2} \det(x, y).$$

On two multiplicity copies use

$$\Omega((x, y), (x', y')) = b(x, y') + b(y, x').$$

Every scalar $a \in \mathbf{F}_4$ is self-adjoint for b , since $b(ax, y) = b(x, ay)$. Thus the vertical line and all four graphs Γ_a are isotropic of half dimension. The five module-theoretic possibilities are exactly the five polarized gluings; no further quadratic condition removes the exotic pair.

The simplex group acts on both multiplicity copies of the coefficient heart over $\mathbf{F}_2$. Hence the three lines defined over $\mathbf{F}_2$ retain S_6 . The stabilizer of an exotic graph is equally intrinsic. A permutation that fixes Γ_ω commutes with ω , hence is $\mathbf{F}_4$ -linear on the two-dimensional $\mathbf{F}_4$ -space H_2 . If its determinant is $d \in \mathbf{F}_4^\times$, preservation of b says

$$\mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(dz) = \mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(z) \quad (z \in \mathbf{F}_4).$$

The trace pairing is nondegenerate, so $d = 1$. It therefore lies in $\mathrm{SL}_2(4)$, of order 60. The given A_5 already has that order and lies in the stabilizer, proving equality. The odd coset in its order-120 normalizer is semilinear and applies field Frobenius, so it exchanges ω and ω^2 .

To pass from the generic fibre to the family, put

$$\mathcal{P} = \mathbf{P}_{\mathbf{F}_4}(\mathbf{F}_4 \otimes_{\mathbf{F}_2} \mathcal{V}_2).$$

By Lemma 2.3, this is the five-sheet local system of A_5 -stable principal halves. It has the degree-three sub-local system $\mathcal{P}_{\mathrm{rat}} = \mathbf{P}_{\mathbf{F}_2}(\mathcal{V}_2)$ and the degree-two complement $\mathcal{P}_{\mathrm{ex}}$. Here H_2 , together with its commutant $\mathbf{F}_4$, is constant, while monodromy acts only on $\mathcal{V}_2$. Its action is therefore $\mathbf{F}_4$ -linear, not semilinear, and factors through $\mathrm{GL}_2(\mathbf{F}_2)$; in particular it preserves both subsets. Identify $\mathrm{GL}_2(\mathbf{F}_2)$ with the permutation group of $\mathbf{P}^1(\mathbf{F}_2)$. Its scalar extension lies in $\mathrm{PGL}_2(\mathbf{F}_4) \simeq A_5$, so the permutation on the complementary pair has the same sign as that on the rational triple. The latter sign defines the orientation cover. Thus the two quadratic classes agree; their torsors are isomorphic in two ways, exchanged by the deck involution. The relative kernel $\mathcal{K}_2$ gives a section of $\mathcal{P}$.

At the geometric generic point this section cannot be rational. Otherwise the natural faithful S_6 -action on $\mathcal{A} = \mathcal{E} \otimes (\mathbf{Z}^\Omega/\mathbf{Z}\mathbf{1})$, which is the identity on $\mathcal{E}$ tensored with the faithful permutation action on the quotient lattice, would preserve the rational kernel: under (2.2) it acts on H_2 and trivially on $\mathcal{V}_2$, so every member of $\mathcal{P}_{\mathrm{rat}}$ is S_6 -stable. At three, Lemma 2.1 says that the analogous coefficient heart H_3 is simple and $\mathrm{End}_{A_5}(H_3) = \mathbf{F}_3$. The semisimple module $H_3^{\oplus 2}$ has stable halves parametrized by $\mathbf{P}^1(\mathbf{F}_3)$. The tensor pairing is b_3 times the alternating multiplicity pairing, so the three affine scalar graphs and the vertical line are all maximal isotropic. The natural S_6 -action is the same on both multiplicity copies, so it fixes every one of these four lines. The dependency is thus: a rational two-primary half is S_6 -fixed, every three-primary half is S_6 -fixed, and $|\ker f| = 6^4$ makes the two primary parts exhaust $\ker f$. The full kernel would therefore be S_6 -stable, so each $s \in S_6$ would descend to $\bar{s} : J \rightarrow J$. The descended map preserves the principal polarization. Indeed s preserves $6I_6 - J_6$, hence $s^*\lambda_{\mathcal{A}} = \lambda_{\mathcal{A}}$, and

$$f^*(\bar{s}^*\lambda_\Theta) = s^*f^*\lambda_\Theta = s^*\lambda_{\mathcal{A}} = \lambda_{\mathcal{A}} = f^*\lambda_\Theta,$$

while $\varphi \mapsto f^\vee \varphi f$ is injective on $\mathrm{Hom}(J, J^\vee)$ because f is an isogeny; so $\bar{s}^*\lambda_\Theta = \lambda_\Theta$ and $\bar{s} \in \mathrm{Aut}(J, \Theta)$. Everything here is a statement about polarization classes, so the translation and Appell–Humbert semicharacter choices that realize a class by an actual line bundle do not enter. The action would be faithful: if a source automorphism acts trivially on the

quotient, the image of its difference from the identity is both connected and contained in the finite isogeny kernel, hence zero. Strong Torelli gives an exact sequence [7, Theorem 3.1]

$$1 \longrightarrow \{\pm 1\} \longrightarrow \text{Aut}(J, \Theta) \longrightarrow \text{Aut}(X) \longrightarrow 1.$$

The kernel of the resulting map from S_6 to $\text{Aut}(X)$ would be a normal subgroup of order at most two; it is trivial. Thus S_6 would act faithfully on a smooth cubic threefold. This contradicts the classification of faithful automorphism groups [7, Theorem 2.1]. Five of the six listed ambient groups have order smaller than $|S_6| = 720$. The remaining group is $(\mathbf{Z}/3\mathbf{Z})^4 \rtimes S_5$, of order 9720, which is not divisible by 720. None can contain S_6 . The section is therefore generically in $\mathcal{P}_{\text{ex}}$. Its inverse image is open and closed in the connected base B° , so every smooth fibre has exotic two-primary kernel. The section therefore trivializes the sign torsor over B° , so the preceding calculation gives $\rho_2(\pi_1(B^\circ)) \subseteq \ker(\text{sgn}) = A_3$. The same calculation at three gives the four lines parametrized by $\mathbf{P}^1(\mathbf{F}_3)$; choosing any complementary line as the vertical ruling makes the selected half an affine scalar graph. $\square$

On a Weierstrass chart $y^2 = x^3 + Ax + B$, the rational packet is the root cover of $x^3 + Ax + B$, while its orientation class is $u^2 = -4A^3 - 27B^2$. Since $\Delta_{\mathcal{E}} = 16(-4A^3 - 27B^2)$, this is the square class of the elliptic discriminant. Invariantly, for the Hodge line λ , it is the class of the torsor of square roots in $\lambda^{\otimes 6}$ of the nowhere-vanishing discriminant section in $\lambda^{\otimes 12}$.

The companion paper [12] identifies this packet with the root and sign quotients of the S_3 -torsor on $\mathcal{E}[2]$ and gives its exact level-six modular realization; those refinements are not used below.

For the cycle-theoretic argument, the unordered pair is enough. Both members act on the heart with the same squarefree minimal polynomial $t^2 + t + 1$. Theorem 3.5 uses only the conclusion that the geometric kernel is not one of the three $\mathbf{F}_2$ -rational gluings. At three it uses only scalarity of the possible A_5 -stable graphs, not the finer monodromy selection.

Hartlieb proves that the closure of its intermediate-Jacobian image is a one-dimensional special subvariety [7, Proposition 5.7]. Since the cubic period map is a locally closed embedding, the resulting smooth family is non-isotrivial. Only this landing and nonconstancy are used below; no identification of compactified parameter curves is needed for the separation theorem.

The exotic kernel also limits how J can be assembled from lower-dimensional principally polarized pieces. Voisin's components of codimension at most three are produced by an odd-degree isogeny from the Jacobian of a possibly reducible curve [14, proof of Theorem 4.5]. A principally polarized abelian variety of dimension at most three is such a Jacobian, so the route is automatic whenever the intermediate Jacobian receives an odd-degree isogeny from a product of factors of dimension at most three under which the polarization pulls back to an odd multiple of the product polarization. Here it is not available.

Proposition 2.5 (No elliptic-product route). *Let J be the intermediate Jacobian of the geometric generic fibre of the nonstandard A_5 -pencil, let E be the geometric generic fibre of $\mathcal{E}$, and let*

$$\mu : \prod_{i=1}^k (A_i, \Theta_i) \longrightarrow (J, \Theta)$$

be an isogeny from a product of principally polarized abelian varieties with $\mu^\Theta = m \sum_i \Theta_i$; its degree is m^5 , since $\deg \lambda_{\mu^*\Theta} = (\deg \mu)^2 \deg \lambda_\Theta$ while $\deg \lambda_{m \sum_i \Theta_i} = m^{10}$, so the degree is odd exactly when m is. If m is odd then either $k = 1$, or $k = 2$ and the two factors have dimensions one and four. The second case occurs: every axis $H \in \Omega$ gives such a μ , with*

A_1 an elliptic curve isogenous to E . Finally, no product of five elliptic curves admits an odd-degree isogeny onto J at all, whatever the pullback of Θ , even though J is isogenous to E^5 .

Proof. Write $\Lambda = \mathbf{Z}^\Omega / \mathbf{Z}\mathbf{1}$ with the symmetric form κ of matrix $6I_6 - J_6$ from Proposition 2.2, put $M = H_1(E, \mathbf{Z})$ and $L = H_1(J, \mathbf{Z})$, and let $\Lambda_1 = \Lambda^\vee \cap \frac{1}{2}\Lambda$, where $\Lambda^\vee$ is the κ -dual. Doubling identifies Λ_1/Λ with the kernel of κ modulo two inside $\Lambda/2\Lambda$, equivariantly, hence with the two-primary coefficient heart H_2 of (2.2); the $\mathbf{F}_4$ -scalars and the pairing b are transported along that identification. By Lemma 2.3, f_* identifies $H_1(\mathcal{A}, \mathbf{Z})$ on the geometric generic fibre with $\Lambda \otimes M \subseteq L$, with quotient $\ker f$, and the polarization form is the tensor product of κ with the symplectic form of M . Fix a basis of $M \otimes \mathbf{Z}_2$. By Proposition 2.4 the two-primary kernel is one of the two exotic graphs, so in that basis

$$L \otimes \mathbf{Z}_2 = \{(u, u') \in (\Lambda_1 \otimes \mathbf{Z}_2)^{\oplus 2} : [u'] = \omega^{\pm 1}[u]\},$$

where $[\cdot]$ is the class in H_2 and ω is the $\mathbf{F}_4$ -scalar of order three.

At the geometric generic point E has no complex multiplication, so the endomorphism algebra of the weight-one Hodge structure $M_{\mathbf{Q}}$ is $\mathbf{Q}$. As $H_1(J, \mathbf{Q}) = W_5 \otimes M_{\mathbf{Q}}$, every sub-Hodge structure is $U \otimes M_{\mathbf{Q}}$ for a subspace $U \subseteq W_5$, and two of them are orthogonal exactly when the subspaces are κ -orthogonal.

Let $U_i \subseteq W_5$ be the subspace with

$$\mu_* H_1(A_i, \mathbf{Q}) = U_i \otimes M_{\mathbf{Q}},$$

and put $L' = \mu_*(\bigoplus_i H_1(A_i, \mathbf{Z}))$. It is the direct sum of sublattices of the sub-Hodge structures $L \cap (U_i \otimes M_{\mathbf{Q}})$, and its index in L is odd because the degree of μ is, so localizing at two turns those inclusions into equalities: $L \otimes \mathbf{Z}_2 = \bigoplus_i P_i$ with $P_i = (L \cap (U_i \otimes M_{\mathbf{Q}})) \otimes \mathbf{Z}_2$. Under the polarization hypothesis the subspaces U_i are moreover κ -orthogonal, since $\mu^* \Theta = m \sum_i \Theta_i$ makes the polarization form block diagonal on $\bigoplus_i H_1(A_i, \mathbf{Z})$; the decomposition $L \otimes \mathbf{Z}_2 = \bigoplus_i P_i$ is then orthogonal and each P_i unimodular, and each summand of L' carries m times a unimodular alternating form. Apply the displayed description of $L \otimes \mathbf{Z}_2$ in two ways. Its first-coordinate projection is all of $\Lambda_1 \otimes \mathbf{Z}_2$ and carries P_i into $(\Lambda_1 \cap U_i) \otimes \mathbf{Z}_2$; summing over i and comparing with that surjectivity gives $\Lambda_1 \otimes \mathbf{Z}_2 = \bigoplus_i (\Lambda_1 \cap U_i) \otimes \mathbf{Z}_2$, with equality in each piece. Its intersection with the first coordinate subspace is $\Lambda \otimes \mathbf{Z}_2$, and that subspace is the direct sum of its intersections with the $U_i \otimes M_{\mathbf{Q}}$, so $\Lambda \otimes \mathbf{Z}_2 = \bigoplus_i (\Lambda \cap U_i) \otimes \mathbf{Z}_2$. Hence $H_2 = \bigoplus_i H_2^{(i)}$ with $H_2^{(i)} = (\Lambda_1 \cap U_i)/(\Lambda \cap U_i)$. For $(u, u') \in P_i$ both classes lie in $H_2^{(i)}$ and, by that same projection, u runs over all of $\Lambda_1 \cap U_i$; so $\omega H_2^{(i)} \subseteq H_2^{(i)}$ and every summand is an $\mathbf{F}_4$ -subspace.

That decomposition is orthogonal for the trace-determinant pairing b used above, and H_2 has $\mathbf{F}_4$ -dimension two, so at most two summands are nonzero, with $\mathbf{F}_4$ -dimensions 2 and 0, or 1 and 1. The second pattern is impossible. An $\mathbf{F}_4$ -line $\ell = \mathbf{F}_4 x$ is totally isotropic for b , since $\det(x, ax) = a \det(x, x) = 0$, and b is nondegenerate, so $\ell^\perp = \ell$; two $\mathbf{F}_4$ -lines orthogonal to one another therefore coincide, and cannot be complementary. One summand consequently carries the whole heart. The discriminant group of a lattice of rank r needs at most r generators and H_2 needs four, so that summand has $\dim U_i \geq 4$. This leaves $k = 1$, or $k = 2$ with dimensions four and one.

For the second case let $v \in \Lambda$ be the class of an axis, so $\kappa(v, v) = 5$. As 5 is a unit at two, $\Lambda \otimes \mathbf{Z}_2$ is the orthogonal sum of $\mathbf{Z}_2 v$ and $(\Lambda \cap v^\perp) \otimes \mathbf{Z}_2$, and the first summand is unimodular, so the whole coefficient heart is carried by the second. By the same display, $L \otimes \mathbf{Z}_2$ is then the orthogonal sum of the localizations of $L \cap (\mathbf{Q}v \otimes M_{\mathbf{Q}})$ and $L \cap (v^\perp \otimes M_{\mathbf{Q}})$, and both are unimodular, so the index of that sum in L is odd. On each summand the alternating form

has odd elementary divisors $d_1 \mid \cdots \mid d_g$; passing to the sublattice spanned by $d_g e_i / d_i$ and f_i in a symplectic basis makes it d_g times a unimodular form, at the odd index $\prod_i (d_g / d_i)$. Taking m to be the product of the two resulting multipliers and shrinking each summand again by the complementary odd factor makes both forms m times a unimodular one. The first summand has rank two and gives an elliptic curve isogenous to E ; the second has rank eight.

For the last assertion drop the polarization hypothesis and let $\mu : \prod_{i=1}^5 E_i \rightarrow J$ be any isogeny of odd degree from a product of elliptic curves. The argument above through $H_2 = \bigoplus_i H_2^{(i)}$ with each summand ω -stable uses only that the index is odd – neither the orthogonality of the P_i nor their unimodularity enters it – so it applies. Each U_i is now a line, so the decompositions $\Lambda_1 \otimes \mathbf{Z}_2 = \bigoplus_i (\Lambda_1 \cap U_i) \otimes \mathbf{Z}_2$ and $\Lambda \otimes \mathbf{Z}_2 = \bigoplus_i (\Lambda \cap U_i) \otimes \mathbf{Z}_2$ force $\Lambda_1 \cap U_i$ and $\Lambda \cap U_i$ to have rank one, and $H_2^{(i)} = (\Lambda_1 \cap U_i) / (\Lambda \cap U_i)$ is therefore cyclic; it is killed by two because $\Lambda_1 \subseteq \frac{1}{2}\Lambda$, and being $\mathbf{F}_4$ -stable it has even dimension over $\mathbf{F}_2$, hence vanishes. Then $H_2 = 0$, against $|H_2| = 16$. $\square$

Hartlieb records, in the footnote to [7, Remark 5.8], that van Geemen and Yamauchi split the intermediate Jacobian of a cubic threefold with an automorphism of order five, up to isogeny, as an elliptic curve times the square of an abelian surface. The proposition is compatible with that: their splitting is an isogeny, and what fails here is its realization by an odd-degree isogeny matching the polarizations. Two routes into Voisin’s construction therefore remain, one for each case of the proposition: a four-dimensional factor of the second case may be the Jacobian of an irreducible curve of genus four, or J itself may be odd-degree isogenous to the Jacobian of an irreducible curve of genus five. Both are open, so this pencil is not known to lie in one of those components. Remark 2.6 says which principally polarized fourfolds the first route ranges over, which is not immediate: the second case admits six of them, and the intrinsic factor they are isogenous to is not one. Section 3 does not depend on the question: the algebraicity of the primitive minimal class is proved there directly.

Remark 2.6 (Which fourfolds the second case admits). Fix an axis H with class v , so that $\kappa(v, v) = 5$, and let $N = v^\perp \subset \Lambda$. In the basis of the five remaining axes, $N = \{x \in \mathbf{Z}^5 : \sum x_i = 0\}$ carries the form $6\langle x, y \rangle$: six times the A_4 root lattice. Since 5 is a unit at two and at three, Λ splits orthogonally as $\mathbf{Z}v \perp N$ over $\mathbf{Z}_2$ and over $\mathbf{Z}_3$, so the whole discriminant group of Λ is carried by N and the kernel of Proposition 2.4 lies in $N \otimes M$. What separates the two summands is five-adic: $[\Lambda : \mathbf{Z}v + N] = 5$, and tensoring with the rank-two lattice M squares that to $[\Lambda \otimes M : (\mathbf{Z}v \otimes M) \oplus (N \otimes M)] = 25$. Passing from $\Lambda \otimes M$ to L enlarges the N -summand alone, by the previous sentence, so this index is also the index in L used below.

Write $L_1 = L \cap (\mathbf{Q}v \otimes M_{\mathbf{Q}})$ and $L_4 = L \cap (N_{\mathbf{Q}} \otimes M_{\mathbf{Q}})$ for the two primitive summands, so that A_1 and the intrinsic four-dimensional factor of the second case have these as their integral homology. Then $L_1 = \mathbf{Z}v \otimes M$ carries 5 times a unimodular alternating form, so $\det L_1 = 25$; since L is unimodular and $[L : L_1 \oplus L_4] = 25$, we get $\det L_1 \cdot \det L_4 = 25^2$ and hence $\det L_4 = 25$. A rank-eight alternating lattice of determinant 25 has elementary divisors $1, 1, 1, 1, 1, 5, 5$, that is polarization type $(1, 1, 1, 5)$, so L_4 is not unimodular and the intrinsic four-dimensional factor is not principally polarized. Its discriminant group is glued to that of L_1 , which is $M/5M = E[5]$; the discriminant pairing is alternating, so every subgroup of order five is isotropic, and each of the six gives a unimodular overlattice of L_4 . The second case of the proposition therefore admits at most six principally polarized abelian fourfolds, each five-isogenous to the intrinsic factor, one for each order-five subgroup of $E[5]$, and exactly six unless two of the overlattices are isomorphic as polarized varieties. The stabilizer D_5 of the axis acts trivially on disc L_4 , since it acts trivially on M and the

gluing is equivariant, so it fixes each of the six and the A_5 -structure selects none of them. The factors the proof of the proposition constructs are obtained from these by a further shrink of odd index, so they are odd-degree isogenous to them rather than equal to them; a genus-four question about the second case is a question about this finite list.

3 Integral divisor products and the primitive minimal class

Let (A, Θ) be a principally polarized abelian variety of dimension g . Put

$$P_A^k = \text{im} \left(\text{Sym}^k \text{NS}(A) \longrightarrow H^{2k}(A, \mathbf{Z}) \right), \quad \gamma_A = \frac{\Theta^{g-1}}{(g-1)!}.$$

The class $D^k/k!$ of a divisor on an abelian variety is integral. The question here is whether these divided powers already belong to the smaller lattice P_A^k generated by ordinary products of integral divisors.

For quotients of elliptic powers, this question is controlled by the valuation pattern of the descended coefficient forms. The main local result of this section says that finite-étale graph slopes force all divided powers of the prescribed graph divisor lattice into ordinary products, in every degree. The proof has two steps: the graph equations produce a symmetric matrix-of-ideals lattice, and a midpoint inequality makes that lattice rank-one generated. Rank-one coefficient divisors have square zero, so their divided powers expand without factorials. The six-axis application supplies the primitive minimal class needed below.

3.1 A local graph theorem

Work over a complete unramified discrete valuation ring O , with uniformizer p . Fix an elliptic curve E , an isogeny $q : E^n \rightarrow Y$ to a principally polarized abelian variety, and extend the p -adic coefficient lattices to O . A symmetric matrix $A \in \text{Sym}_n(O)$ denotes the standard coefficient divisor D_A on E^n , generated by coordinate and Poincare divisors. Define the prescribed graph-divisor lattice and its ordinary product image by

$$N_{\text{gr}}(q, O) = \{D \in \text{NS}(Y) \otimes O : q^*D = D_A \text{ for some } A \in \text{Sym}_n(O)\},$$

$$P_{\text{gr}}^k = \text{im} \left(\text{Sym}^k N_{\text{gr}}(q, O) \longrightarrow H^{2k}(Y, \mathbf{Z}_p) \otimes O \right).$$

Because q^* is injective on torsion-free cohomology, we identify $N_{\text{gr}}(q, O)$ with the symmetric matrices whose coefficient divisors descend through q . This is the prescribed lattice even when E has complex multiplication; no extra CM divisor is included.

Suppose the pullback coefficient polarization on the elliptic-power source is

$$G = PB, \quad P = \bigoplus_a p^a I_{M_a}, \quad B = \bigoplus_a B_a,$$

where each B_a is unimodular. A *marked finite-étale graph presentation* consists of a symplectic ruling of the primary polarization kernel in which $\ker q$ is the graph of an endomorphism T , with the following properties: T preserves the depth summands M_a ,

$$T^t B^{-1} = B^{-1} T \quad \text{or, equivalently,} \quad B T^t = T B, \quad (3.1)$$

and the algebra generated by T modulo p^a is finite étale on every positive-depth summand. Thus adjointness is taken for the dual coefficient form B^{-1} . No étaleness is required on M_0 .

After a finite unramified extension splitting these étale algebras, the positive-depth summands decompose orthogonally into blocks

$$L_i = L_{a_i, \lambda_i}, \quad T_i = t_i I + p^{a_i} S_i, \quad (3.2)$$

with S_i integral. Indeed, completeness lifts the primitive idempotents of each split finite-étale quotient to idempotents in the algebra generated by T . They are polynomials in T , so (3.1) makes them self-adjoint for B^{-1} . Their images are therefore mutually orthogonal for B^{-1} , and hence for B ; on each image, T is scalar modulo the corresponding power p^{a_i} . We keep the unit part as one block. Write $\delta_{ij} = v_p(t_j - t_i)$, with $v_p(0) = \infty$, for two positive-depth blocks.

Lemma 3.1 (The graph coefficient lattice). *In the split basis, the coefficient lattice of descended graph divisors is the symmetric matrix-of-ideals lattice whose diagonal block i has depth p^{a_i} and whose (i, j) -block has depth*

$$e_{ij} = \max\{a_i, a_j, a_i + a_j - \delta_{ij}\}. \quad (3.3)$$

For the unit block and a positive-depth block j , one has $e_{0j} = a_j$. In particular,

$$e_{ij} \geq \max(a_i, a_j) \geq \left\lceil \frac{a_i + a_j}{2} \right\rceil. \quad (3.4)$$

Proof. Choose the symplectic basis determined by the marked ruling and the depth decomposition. Adjoining the graph generators to the source homology lattice gives the quotient lattice, and in these coordinates the change-of-lattice matrix is

$$C = \begin{pmatrix} P^{-1} & P^{-1}T \\ 0 & I \end{pmatrix}.$$

A coefficient alternating form descends precisely when its matrix transported by C is integral. For a symmetric source coefficient A , direct multiplication gives

$$C \begin{pmatrix} 0 & A \\ -A & 0 \end{pmatrix} C^t = \begin{pmatrix} P^{-1}(AT^t - TA)P^{-1} & P^{-1}A \\ -AP^{-1} & 0 \end{pmatrix}. \quad (3.5)$$

Thus descent is equivalent to the integrality of the three displayed nonzero blocks. Here $A = A^t$ is a symmetric coefficient bilinear form in this normalization. On two positive-depth blocks $L_i \times L_j$, the two off-diagonal blocks first force A_{ij} to have depth $\max(a_i, a_j)$. Substituting $T_i = t_i I + p^{a_i} S_i$ into the upper-left block leaves the additional condition

$$p^{-a_i - a_j}(t_j - t_i)A_{ij} \in M_{r_i \times r_j}(O);$$

the terms containing S_i, S_j are already integral. Intersecting the two conditions gives (3.3). Changing a scalar lift cannot change its maximum. Pairing the unsplit unit block with L_j gives $e_{0j} = a_j$ directly from (3.5), and (3.4) follows. $\square$

The inequality (3.4) is the valuation shadow of the tropical positive semidefinite cone studied by Yu [16]. We need the following signed integral lift of that inequality; in particular, no division by two is permitted.

Lemma 3.2 (Rank-one generation over a DVR). *Let L be a symmetric matrix-of-ideals lattice over O , with diagonal depths p^{a_i} and cross depths $p^{e_{ij}}$. Then L is generated by its rank-one forms if and only if*

$$2e_{ij} \geq a_i + a_j \quad (i \neq j).$$

Proof. If $cvv^t \in L$, the two diagonal conditions imply

$$v_p(cv_i v_j) \geq \left\lceil \frac{a_i + a_j}{2} \right\rceil.$$

Hence rank-one forms cannot generate a shallower cross ideal.

Conversely, fix a generator $p^e(uv^t + vu^t)$ of a cross block. Choose $x \geq a_i$ with $y = 2e - x \geq a_j$. The integers x, y have the same parity. If $t = \min(x, y)$, write $x = t + 2r$ and $y = t + 2s$. Then

$$p^t(p^r u + p^s v)(p^r u + p^s v)^t - p^x u u^t - p^y v v^t = p^e(uv^t + vu^t).$$

All three rank-one terms lie in L . Together with the diagonal rank-one forms, these identities generate L , without division by two. $\square$

Theorem 3.3 (All-degree graph saturation). *For a marked finite-étale graph presentation, every divided power of a class in its prescribed graph divisor lattice is an ordinary integral divisor product. Equivalently, after localization at p , the divided-power envelope of that lattice and its ordinary product image agree in every degree.*

Proof. After the unramified splitting extension, Lemmas 3.1 and 3.2 express every class in the extended graph divisor lattice as a finite sum $D = R_1 + \cdots + R_m$ of rank-one coefficient classes. These are classes in the coefficient extension of the original lattice, not newly descended line bundles. On the elliptic-power source the alternating form of each R_i has rank two, hence $R_i^2 = 0$. Pullback by the isogeny is injective on torsion-free cohomology, so the same identity holds on the quotient. Therefore

$$\frac{D^k}{k!} = \sum_{1 \leq i_1 < \cdots < i_k \leq m} R_{i_1} \cdots R_{i_k}, \quad (3.6)$$

an O' -linear combination of ordinary products over the splitting ring O' . More precisely,

$$P_{\text{gr}}^k \otimes_O O' = \text{im} \left(\text{Sym}^k(N_{\text{gr}} \otimes_O O') \longrightarrow H^{2k} \otimes_O O' \right).$$

Faithful flatness descends the quotient of these finite lattices, and hence descends (3.6) to O , and therefore to $\mathbf{Z}_p$ in the six-axis application. $\square$

The theorem concerns the divisor lattice selected by the marked elliptic-power presentation. If E has no complex multiplication, this is the full Neron–Severi lattice. At a CM fibre extra divisor classes need not satisfy the graph hypotheses. The statement is cohomological: it neither constructs Chow divided powers nor asserts that every integral Hodge class is a divisor product.

For comparison with the classical rational theory, Milne proves that the Lefschetz classes are the rational algebra generated by divisors [9, Theorem 3.2]. Divided powers in Chow and integral Fourier transforms are studied in [10, 2]. The local result above instead keeps the weighted integral graph lattice and the ordinary, rather than divided, product image.

3.2 The six-axis packet

We now apply the graph theorem to the polarization computed in Section 2.

We use the following elementary depth-one lifting observation. Let U be a unimodular O -lattice with symmetric Gram matrix B . Every endomorphism $\bar{T}$ of U/pU that is self-adjoint for $\bar{B}^{-1}$ has a lift $T \in \text{End}_O(U)$ satisfying

$$T^t B^{-1} = B^{-1} T. \quad (3.7)$$

The construction may be performed separately on orthogonal depth summands. Choose any lift T_0 . Its adjointness defect has the form

$$T_0^t B^{-1} - B^{-1} T_0 = pD,$$

where D is skew-symmetric with zero diagonal. Choose an integral matrix C with $C^t - C = -D$, for example by assigning its entries strictly above and below the diagonal. Then $T = T_0 + pBC$ reduces to $\bar{T}$, and

$$T^t B^{-1} - B^{-1} T = p(D + C^t - C) = 0.$$

No division by two is used. Applying the construction blockwise preserves the depth decomposition.

Lemma 3.4 (Local chart for the six-axis quotient). *Let $G = 6I_5 - J_5$. At each prime $p \mid 6$, the coefficient lattice has an orthogonal decomposition*

$$(\mathbf{Z}_p^5, G) \simeq U_0 \perp pU_1,$$

where U_0 has rank one and U_1 rank four, both unimodular. The principal quotient is trivial on U_0 . On U_1/pU_1 , its graph slope has minimal polynomial $t^2 + t + 1$ at $p = 2$; at $p = 3$ it is a scalar, hence a one-point finite-étale block, after choosing a complementary symplectic ruling if the selected half is vertical. In both cases the slope has a depth-preserving integral lift that is self-adjoint for the dual coefficient form, so these data give a marked finite-étale graph presentation.

Proof. Let $e_1, \dots, e_5$ be the standard basis. Since $\langle e_1, e_1 \rangle = 5$ and $\langle e_1, e_j \rangle = -1$, the vectors

$$v_j = e_j + \frac{1}{5}e_1 \quad (2 \leq j \leq 5)$$

span the orthogonal complement of e_1 over $\mathbf{Z}_p$. Its Gram matrix is $\frac{6}{5}(5I_4 - J_4)$, and $\det(5I_4 - J_4) = 5^3$. This gives the asserted unit line and depth-one block at two and three. The unit line contributes nothing to the p -primary discriminant group, so the p -primary isogeny kernel is supported entirely on U_1/pU_1 .

At two, Proposition 2.4 applies to every smooth fibre and identifies its geometric kernel with one of the two exotic graphs. Isotropy says that either slope is self-adjoint for the reduced dual form. It is a root in $\mathbf{F}_4$ of $t^2 + t + 1$, so its generated algebra is finite étale. The preceding lifting argument supplies the required integral self-adjoint lift on U_1 . At three, the same proposition says that every A_5 -stable half is a scalar graph because the coefficient heart is simple with commutant $\mathbf{F}_3$ by Lemma 2.1. More precisely, the four halves form $\mathbf{P}^1(\mathbf{F}_3)$. If the selected member is vertical, choose any other member as the complementary vertical ruling; the selected half then becomes the graph of a scalar (including zero). This change is only in the symplectic ruling of the primary kernel. The chosen symplectic basis of its two-dimensional multiplicity space lifts from $\mathbf{F}_3$ to $\mathbf{Z}_3$: equivalently, reduction $\mathrm{Sp}_2(\mathbf{Z}_3) \rightarrow \mathrm{Sp}_2(\mathbf{F}_3)$ is surjective. Thus it lifts to an integral change of ruling preserving the alternating form; the depth-one coefficient block and its scale $P = 3I$ are unchanged. The algebra generated by the resulting scalar is $\mathbf{F}_3$, hence is a one-point finite-étale block, and the same lifting argument applies. Extend each lift by a scalar on U_0 . No global choice of either slope is needed. $\square$

Theorem 3.5 (Divided powers for the six-axis family). *For every smooth member X_b of the nonstandard A_5 -cubic pencil, the prescribed six-axis graph divisor lattice on J_b has all its integral divided powers in the ordinary divisor-product image. In particular,*

$$\frac{\Theta_b^4}{4!} \in P_{J_b}^4. \quad (3.8)$$

Proof. Only the primes two and three occur in the Smith form $(1, 6, 6, 6, 6)$. Lemma 3.4 verifies the hypotheses of Theorem 3.3 at both primes. At every other prime the isogeny identifies the integral source and target lattices; equivalently the marked presentation has only its unit block $P = I$, so Theorem 3.3 applies trivially (or directly by Lemma 3.2). Thus the image of each desired divided power in the finitely generated quotient $H^{2k}(J_b, \mathbf{Z})/P_{J_b}^k$ vanishes after tensoring with $\mathbf{Z}_p$ for every prime p , and hence vanishes integrally: a finitely generated abelian group splits as a free part and a finite part, and both are detected by these completions. Finally, Lemma 2.3 gives $f_b^* \Theta_b = \lambda_{A_b}$. The principal polarization is therefore one of the descended graph divisors defining the prescribed lattice, yielding (3.8). Extra Neron–Severi classes at a special fibre can only enlarge the ordinary product image needed for this last assertion. $\square$

Voisin identifies the primitive minimal class as the exact obstruction needed for cubic threefolds.

Corollary 3.6 (Universal CH_0 -triviality). *Every smooth member of the nonstandard A_5 -cubic pencil is universally CH_0 -trivial.*

Proof. By Theorem 3.5, the primitive minimal class of the intermediate Jacobian is algebraic. Voisin’s Corollary 4.4 [14] says that, for a smooth complex cubic threefold, this is equivalent to universal CH_0 -triviality. $\square$

Two comparisons place the pencil against the parts of the algebraic locus that were known before. Write M_{H_1} for the coarse-moduli image of the pencil, as in Hartlieb’s notation. Both comparisons run through one classical criterion.

Lemma 3.7 (Eckardt criterion). *Let $X = \{F = 0\} \subset \mathbf{P}^4$ be a smooth cubic threefold and let $p \in X$. The tangent hyperplane section $T_p X \cap X$ is a cone with vertex p – that is, p is an Eckardt point of X – if and only if the Hessian matrix of F at p has rank at most two. Every smooth cubic threefold in either of two classes carries an Eckardt point: those whose defining form is a sum of cubic forms in pairwise disjoint nonempty groups of at most three variables, and the members of the family of [17, Theorem 3.3].*

Proof. Choose coordinates with $p = [1 : 0 : 0 : 0 : 0]$ and write $F = x_0^2 L + x_0 Q + C$ with L, Q and C in $x_1, \dots, x_4$. Smoothness at p makes L nonzero, the tangent hyperplane is $\{L = 0\}$, and the section is a cone with vertex p exactly when Q vanishes on $\{L = 0\}$, that is when $Q \in (L)$. The Hessian of F at p is the matrix of the quadric $4x_0 L + 2Q$. If $Q = LR$ for a linear form R , that quadric is $L(4x_0 + 2R)$, of rank at most two. Conversely a quadric of rank at most two is a product of two linear forms; the coefficient of x_0^2 vanishes, so at most one factor involves x_0 , and if neither did then L would vanish. Comparing the terms linear in x_0 shows that the other factor is proportional to L , so $Q \in (L)$.

Let $F = \sum_j F_j$ be separated, with the j -th form in $n_j \leq 3$ variables and $\sum_j n_j = 5$. Either some $n_j = 2$, or the parts are 3, 1, 1 or 1, 1, 1, 1 and at least two of them are singletons. In the first case the binary cubic F_j has a nonzero root, and the corresponding point of $\mathbf{P}^4$, supported on those two variables, lies on X . In the second, let cx^3 and $c'y^3$ be two singleton summands, with c and c' nonzero because X is smooth, and take the point supported on x and y with coordinates satisfying $ca^3 + c'b^3 = 0$. In both cases every diagonal block of the Hessian belonging to another part vanishes there, since a cubic form has vanishing Hessian at the origin, so the rank is at most two.

For the family of [17, Theorem 3.3] work in their coordinates $x_1, \dots, x_5$ and take p to be the point where $x_1 = 1$ and the other four vanish. It lies on the threefold; every partial derivative of their form vanishes at p except the one in x_3 , whose value is their nonzero

coefficient t_1 , so the tangent hyperplane is $\{x_3 = 0\}$; and the restriction of their form to that hyperplane does not involve x_1 , so the section is a cone with vertex p . $\square$

The general member of the pencil carries no such point, and the members that do are pinned down by a classification rather than by a search. The mechanism is that an Eckardt point is the same object as a reflection preserving the defining form. A cubic threefold with an irreducibly acting automorphism group and an Eckardt point therefore has its form invariant under an irreducible complex reflection group of rank five, and the invariant degrees of those groups leave two possibilities, of which one is singular.

Lemma 3.8 (Eckardt points are reflection centres). *Let $X = \{F = 0\} \subset \mathbf{P}^4$ be a smooth cubic threefold. Call $r \in \mathrm{GL}_5(\mathbf{C})$ a reflection if it has order two and fixes a hyperplane pointwise, and call the line on which it acts by -1 its centre. Sending r to its centre is a bijection from the set of reflections with $F \circ r = F$ onto the set of Eckardt points of X .*

Proof. Let r be a reflection with $F \circ r = F$, and choose coordinates with centre $p = [1 : 0 : 0 : 0 : 0]$ and $r : x_0 \mapsto -x_0$. Invariance kills the odd powers of x_0 , so $F = x_0^2 L + C$ with L and C in $x_1, \dots, x_4$. Then $F(p) = 0$, so $p \in X$; the Hessian of F at p carries the coefficients of L in its first row and column and vanishes elsewhere, so it has rank at most two; and $L \neq 0$, since otherwise every first partial derivative of F would vanish at p . By Lemma 3.7, p is an Eckardt point.

The map is injective. A second reflection r' with $F \circ r' = F$ and the same centre has mirror $\{x_0 = \ell\}$ for a linear form ℓ in $x_1, \dots, x_4$. In the coordinate $u = x_0 - \ell$ the form reads $F = u^2 L + 2u\ell L + (\ell^2 L + C)$, and invariance under $u \mapsto -u$ forces $\ell L = 0$, hence $\ell = 0$ and $r' = r$.

The map is surjective. Let p be an Eckardt point, and in coordinates with $p = [1 : 0 : 0 : 0 : 0]$ write $F = x_0^2 L + x_0 Q + C$ with L, Q, C in $x_1, \dots, x_4$; by the proof of Lemma 3.7, $L \neq 0$ and $Q = LR$ for a linear form R . Substituting $x_0 - R/2$ for x_0 fixes p and turns F into $x_0^2 L + (C - LR^2/4)$, so the reflection $x_0 \mapsto -x_0$ fixes F and has centre p . $\square$

Proposition 3.9 (Irreducible symmetry with an Eckardt point). *Let $X = \{F = 0\} \subset \mathbf{P}^4$ be a smooth cubic threefold whose automorphism group acts irreducibly on $\mathbf{C}^5$. If X carries an Eckardt point, then X is projectively equivalent to the Fermat cubic threefold and carries exactly thirty Eckardt points.*

Proof. Let $R \subset \mathrm{GL}_5(\mathbf{C})$ be generated by the reflections r with $F \circ r = F$; by Lemma 3.8 there is at least one. Every generator fixes F , so F is an R -invariant of degree three, and every reflection of R again fixes F and so belongs to that generating set. The group is finite: its image in $\mathrm{PGL}_5(\mathbf{C})$ lies in the finite group $\mathrm{Aut}(X)$, and each generator has determinant -1 , so a scalar $c \in R$ satisfies $c^5 = \pm 1$.

The action of R is irreducible. Let $\tilde{G}$ be the group of matrices carrying F to a scalar multiple of itself. Conjugating a reflection that fixes F by an element of $\tilde{G}$ gives another, so $\tilde{G}$ normalizes R and permutes the isotypic components of $\mathbf{C}^5$ under R . The sum over an orbit of components is $\tilde{G}$ -invariant, and $\tilde{G}$ surjects onto $\mathrm{Aut}(X)$, so the permutation is transitive and the number of components divides five. Five components would make R diagonal in the resulting basis, its reflections the sign changes in a single coordinate, and transitivity would put all five of those in R ; a cubic form invariant under all five is even in each variable, hence zero. So there is one isotypic component. Its irreducible constituent is not one-dimensional, since then R would act by scalars and contain no reflection. Hence $\mathbf{C}^5$ is irreducible under R , which is therefore an irreducible complex reflection group of rank five.

Three is one of its degrees. The invariants of R form a polynomial ring on five homogeneous generators, each of degree at least two because an irreducible reflection group fixes no linear form. A nonzero invariant of degree three is therefore a scalar multiple of a generator of degree three.

Two groups have one. The irreducible complex reflection groups of rank five are $G(1, 1, 6)$, the groups $G(m, p, 5)$ with $m \geq 2$ and $p \mid m$, and G_{33} ; their degrees are 2, 3, 4, 5, 6, then $m, 2m, 3m, 4m, 5m/p$, and 4, 6, 10, 12, 18 [13]. Three occurs only for $G(1, 1, 6)$ and for $m = 3$, since $5m/p = 3$ would need $p > m$; and in each case it occurs once, so F is that generator up to scalar. For $G(1, 1, 6)$, the symmetric group on six letters acting on the sum-zero subspace of $\mathbf{C}^6$, the generator is the third elementary symmetric function and its zero locus is the Segre cubic threefold, which has ten nodes; that contradicts smoothness. For $G(3, p, 5)$, acting by monomial matrices whose nonzero entries are cube roots of unity, the generator is $y_1^3 + \cdots + y_5^3$. So X is the Fermat cubic threefold in those coordinates.

It remains to count. R is generated by reflections of order two, and the order-two reflections of $G(3, 1, 5)$ generate its subgroup $G(3, 3, 5)$, so $R = G(3, 3, 5)$, whose degrees are 3, 6, 9, 12, 5. The Eckardt points of X correspond to the reflections of R by Lemma 3.8 and the first paragraph, and an irreducible reflection group has $\sum_i (d_i - 1)$ reflections, here $2 + 5 + 8 + 11 + 4 = 30$. $\square$

Remark 3.10. The argument does not use the dimension until the last two steps, and running it for a smooth cubic hypersurface in $\mathbf{P}^N$ with $N \geq 3$ leaves the Fermat cubic together with the diagonal cubic $\{e_1 = e_3 = 0\}$, the invariant of the symmetric group on $N + 2$ letters acting on the sum-zero subspace of $\mathbf{C}^{N+2}$. The diagonal cubic is singular exactly when N is even: a singular point forces all x_i^2 equal, hence coordinates $\pm t$ in equal numbers. For $N = 3$ both survive, as the Fermat and Clebsch cubic surfaces, with $\sum_i (d_i - 1)$ equal to eighteen and ten reflections; those are their classical Eckardt counts. For $N = 4$ the diagonal cubic is the Segre cubic threefold and only the Fermat member survives, which is the case used above. The restriction $N \geq 3$ is needed: for plane cubics the group $G(3, 3, 3)$ has degree three twice, its invariants of that degree are the Hesse pencil, and every smooth plane cubic accordingly has inflection points.

Proposition 3.11 (The Eckardt locus of the pencil). *Exactly two members of the pencil are smooth cubic threefolds carrying an Eckardt point. They are interchanged by the normalizer of A_5 in $\mathrm{PGL}(W_5)$, each carries exactly thirty Eckardt points, and both are projectively equivalent over $\mathbf{C}$ to the Fermat cubic threefold, so they represent a single point of the coarse-moduli image M_{H_1} . Every other point of M_{H_1} is represented only by smooth cubic threefolds with no Eckardt point; in particular no other point is represented by a cubic projectively equivalent to a member of the family of [17, Theorem 3.3].*

Proof. Every member of the pencil is invariant under A_5 acting irreducibly on W_5 , so Proposition 3.9 applies to each member of B° : one carrying an Eckardt point is projectively equivalent to the Fermat cubic threefold and carries exactly thirty. Conversely the pencil contains the Fermat cubic threefold by Remark 1.2, and the Fermat form is separated, so Lemma 3.7 gives it an Eckardt point. The members carrying an Eckardt point are therefore exactly those projectively equivalent to the Fermat cubic threefold, and they represent the single point of M_{H_1} that it represents.

Counting them uses two facts about the ambient group. Write A for the image of A_5 in $\mathrm{PGL}(W_5)$ and N for its normalizer. The centralizer of A consists of scalars by Schur's lemma, so conjugation embeds N in $\mathrm{Aut}(A_5) \cong S_5$ with A as the inner automorphisms; the six-point model realizes the outer automorphism, since $\mathrm{PGL}(2, \mathbf{F}_5)$ permutes the six points of $\mathbf{P}^1(\mathbf{F}_5)$ and so acts on W_5 . Hence $N \cong S_5$ and $[N : A] = 2$. Second, any two subgroups

of $\mathrm{PGL}_5(\mathbf{C})$ isomorphic to A_5 and acting irreducibly are conjugate: a five-dimensional projective representation of A_5 is linear, because the faithful representations of its double cover have dimensions two, four and six; A_5 has one irreducible character of degree five; and that character is fixed by $\mathrm{Aut}(A_5)$.

Write X_F for the Fermat cubic threefold. Sending a parameter b with X_b projectively equivalent to X_F to the $\mathrm{Aut}(X_F)$ conjugacy class of gAg^{-1} , for any g with $gX_b = X_F$, is well defined, since two such g differ by an element of $\mathrm{Aut}(X_F)$. It is surjective onto the classes of irreducibly acting copies of A_5 in $\mathrm{Aut}(X_F)$: given such a copy A' , conjugacy gives h with $hAh^{-1} = A'$, and $h^{-1}X_F$ is A -invariant, hence a member of the pencil mapping to the class of A' . Its fibres are the orbits of N : if $u \in \mathrm{Aut}(X_F)$ conjugates gAg^{-1} to $g'A(g')^{-1}$ then $n = (g')^{-1}ug$ normalizes A and carries X_b to $X_{b'}$, and conversely $n \in N$ with $nX_b = X_{b'}$ turns g into gn^{-1} .

Now $\mathrm{Aut}(X_F)$ is $C_3^4 \rtimes S_5$, the diagonal cube roots of unity modulo scalars extended by the coordinate permutations [15, Example 3.1]. A subgroup isomorphic to A_5 meets C_3^4 in one of its own normal abelian subgroups, hence trivially, so it is a complement to C_3^4 over the alternating subgroup, and such complements are classified up to C_3^4 -conjugacy by $H^1(A_5, C_3^4)$. As a module, C_3^4 is the sum-zero submodule of $\mathbf{F}_3^5$, which splits off the diagonal line because five is invertible modulo three; A_5 has no homomorphism to $\mathbf{F}_3$; and $\mathbf{F}_3^5$ is induced from the trivial module of a point stabilizer A_4 . So Shapiro's lemma gives

$$H^1(A_5, C_3^4) = H^1(A_5, \mathbf{F}_3^5) = H^1(A_4, \mathbf{F}_3) = \mathrm{Hom}(A_4, \mathbf{F}_3) \cong \mathbf{F}_3.$$

The trivial class is the group of coordinate permutations, which acts reducibly. An element of $\mathrm{Aut}(X_F)$ lying over a transposition acts on $H^1(A_5, C_3^4)$ by inversion, because conjugation by a transposition inverts the cyclic quotient of A_4 , so the two remaining classes are fused. Exactly one class of irreducibly acting copies of A_5 survives, and the parameters with X_b projectively equivalent to X_F form a single N -orbit.

That orbit has two elements. If N fixed one of them, then N , which is isomorphic to S_5 and acts irreducibly, would sit inside $\mathrm{Aut}(X_F)$; the same argument would make it a complement to C_3^4 , and $H^1(S_5, C_3^4) = H^1(S_4, \mathbf{F}_3) = 0$ leaves only the group of coordinate permutations, which acts reducibly. So N permutes the two members without fixing either, and $[N : A] = 2$ makes them a single N -orbit of size two.

Finally every member of the family of [17, Theorem 3.3] carries an Eckardt point, again by Lemma 3.7, and carrying one is invariant under projective equivalence, which gives the last claim. $\square$

The two members can be named in coordinates, and the pencil's singular members with them. Nothing above or below depends on this.

Lemma 3.12 (The two loci in the six-point model). *Write the pencil as $p_3 + bT_1$ in the six-point model of §2, in the chart where the coefficient of p_3 is one. The two members of Proposition 3.11 are the roots of $b^2 + 3b + 9$, that is $b = 3\omega$ for a primitive cube root of unity ω . The singular members are the roots of $b(b+6)(b^2 - 3b - 9)(7b^2 + 3b + 9)$, the first of them the Segre cubic threefold. The remaining pencil point, the cubic T_1 , is smooth with no Eckardt point.*

Proof. By Lemma 3.7 a point p of a smooth X_b is an Eckardt point exactly when the Hessian of F_b at p has rank at most two, so the parameters sought are those for which the ideal generated by F_b and the three-by-three minors of its Hessian has a zero away from the origin. Saturating that ideal by the irrelevant ideal and eliminating the five coordinates gives $b(b^2 + 3b + 9)$; the same for the Jacobian ideal alone gives $b(b+6)(b^2 - 3b - 9)(7b^2 + 3b + 9)$, whose zero set is the reduced locus of singular parameters, that is the radical of the pencil's

discriminant rather than the discriminant itself. Both are Gröbner eliminations over $\mathbf{Q}$, trusted as executions of a computer algebra system rather than certificate-checked; the registered evidence bundle records the replay commands and two independent cross-checks. They are carried out in the chart above, and the remaining pencil point is treated directly there.

Elimination computes the ideal of the Zariski closure of the projection, so this identifies the set of parameters sought only up to closure. Here the two agree: the set is constructible by Chevalley's theorem, its closure is finite, and a constructible subset of the affine line with finite closure is itself finite, hence closed.

The factor b common to the two eliminated ideals is the Segre cubic threefold, the second branch of Proposition 3.9, which is singular, so it is not a member of B° and Lemma 3.7 does not apply to it. The roots of $b^2 + 3b + 9$ are prime to the second polynomial, so those two members are smooth and carry an Eckardt point; Proposition 3.11 shows that these are the only such members. $\square$

Proposition 3.13 (The separated-variable intersection). *Exactly one point of M_{H_1} is represented by a smooth cubic threefold that is projectively equivalent to a cubic whose defining form is a sum of cubic forms in pairwise disjoint nonempty groups of at most three variables, namely the Fermat point of Proposition 3.11. Consequently, every other point of M_{H_1} is represented by universally CH_0 -trivial cubic threefolds by Corollary 3.6, none of which is covered by the separated-variable criterion of Colliot-Thélène [4].*

Proof. By Lemma 3.7 a smooth cubic threefold of separated-variable type carries an Eckardt point, and carrying one is invariant under projective equivalence, so the points of M_{H_1} represented by such cubics are among those of Proposition 3.11, which is the Fermat point alone. That point is represented by such a cubic: the Fermat form $x_1^3 + \cdots + x_5^3$ is a sum of cubic forms in five disjoint groups of one variable each. Hence the two loci coincide. Every remaining point is represented by a universally CH_0 -trivial cubic threefold by Corollary 3.6, and by no cubic projectively equivalent to a separated-variable form, so the criterion of [4] does not apply to it. $\square$

The order-three signatures give finiteness, though not the exact locus, by an independent route. Fix a primitive cube root of unity ζ_3 . The signature of an order-three automorphism of a smooth cubic threefold is the tuple of exponents of ζ_3 in a diagonalization of a linear lift of order three; it is well defined up to permutation of the entries, replacement of ζ_3 by ζ_3^2 , and multiplication of the lift by a cube root of unity [5, Definition 2.1]. In a separated-variable form, as in Proposition 3.13, the groups of variables are disjoint, nonempty and of size at most three, and they cover all five variables, since a cubic threefold whose form omits a variable is singular at the corresponding coordinate point; as five is not a multiple of three, at least one group has size one or two. Multiplying the variables of such a group by ζ_3 fixes the form, so such a cubic admits an order-three automorphism of signature $(0, 0, 0, 0, 1)$ or $(0, 0, 0, 1, 1)$. González-Aguilera–Liendo identify these two signatures as their families T_3^1 and T_3^2 [5, Theorem 2.5]. The union of the two families therefore contains every point of M_{H_1} represented by a cubic projectively equivalent to a separated-variable form. Each family is the image in moduli of $U^G/N_{\mathrm{GL}_5(\mathbf{C})}(G)$, for U^G the locus of smooth cubics fixed by the relevant group G and $N_{\mathrm{GL}_5(\mathbf{C})}(G)$ its normalizer, under a finite morphism, hence closed [7, Lemmas 2.6–2.7]. The Klein cubic lies in the pencil [7, Lemmas 2.6–2.7 and Proposition 5.7], its automorphism group $\mathrm{PSL}(2, \mathbf{F}_{11})$ has a single conjugacy class of elements of order three, and its signature is $(0, 0, 1, 1, 2)$ [5, Proposition 2.6]. The multiplicities of the eigenvalues $1, \zeta_3, \zeta_3^2$ are $(2, 2, 1)$ for that signature and $(4, 1, 0)$ and $(3, 2, 0)$ for the two signatures above. Permuting the entries leaves these triples unchanged, multiplying the lift by a cube root of

unity permutes them cyclically, and replacing ζ_3 by ζ_3^2 transposes the last two, so the three moves compare the triples only as multisets; as $\{1, 2, 2\}$, $\{0, 1, 4\}$ and $\{0, 2, 3\}$ are distinct, the Klein cubic lies in neither family. Some $b \in B^\circ$ has X_b projectively equivalent to the Klein cubic, by the membership just recorded, and B° is a nonempty open subset of a $\mathbf{P}^1$, hence an irreducible curve. The preimage in B° of each of the two loci under the map to moduli is closed and omits that b , so it is a proper closed subset of B° and therefore finite. Since M_{H_1} is the image of B° , each locus meets M_{H_1} in a finite set. This is a second proof of finiteness, from prime-order signatures rather than from reflections, and it does not identify the finite set; that identification is Propositions 3.11 and 3.13.

Every moduli point of the pencil but the Fermat point escapes the separated-variable mechanism of [4] by Proposition 3.13 and the explicit family of [17] by Proposition 3.11, and Proposition 2.5 closes the product route into Voisin’s components. None of them asserts that the A_5 -curve is disjoint from every known part of Voisin’s algebraic-minimal-class locus; the two questions left open there are recorded after Proposition 2.5.

The argument is fibrewise. It constructs neither a horizontal minimal cycle over the pencil nor a relative decomposition of the diagonal. Neither is needed for Corollary 3.6.

4 The family consequence

Proof of Theorem 1.1. Corollary 3.6 gives universal CH_0 -triviality of $\mathcal{X}_b$, and the projective-bundle formula gives the same property for $\mathcal{X}_b \times \mathbf{P}^1$. The one-stabilization theorem [18, Theorem 1.1] proves that this fourfold is irrational.

Non-isotriviality follows from the nonconstant intermediate-Jacobian period map on the A_5 -component: Hartlieb proves that the closure of its image is a one-dimensional special subvariety [7, Proposition 5.7]. Finally, Proposition 3.13 shows that every point of this moduli curve except the Fermat point lies outside the separated-variable locus. $\square$

The six-axis mechanism is independent of the irrationality theorem it is paired with. It constructs the algebraic primitive minimal class uniformly along the pencil; the QDM marker of [18] applies to every smooth cubic threefold. Their conjunction gives the separation asserted above.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author’s direction, the systems assisted with proof exploration, literature research, exact computation, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

Numbered statements are cited from the exact versions listed here. Where both a journal and a preprint version appear, the numbering is that of the preprint.

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